

WEEK-3 PRACTICAL

Task 3: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

CODE:

```
GNU nano 8.7.1 ParentChil
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("State      : Running\n\n");

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child process created\n");
        printf("Child PID   : %d\n", getpid());
        printf("Child PPID  : %d\n", getppid());
        printf("State      : Running\n");

        sleep(3);

        printf("Child PID %d\n", getpid());
        printf("State      : Terminated\n");

        exit(0);
    }
}
```

```

        exit(0);
    }
    else
    {
        printf("\nParent process\n");
        printf("Parent PID   : %d\n", getpid());
        printf("Child PID    : %d\n", pid);
        printf("State       : Waiting\n");

        wait(NULL);

        printf("Child process completed\n");
        printf("Parent State: Running\n");
        printf("Parent State: Terminated\n");
    }

    return 0;
}

```

OUTPUT:

```

saicharan_reddi@SAICHARANREDDY:~$ nano ParentChild.c
saicharan_reddi@SAICHARANREDDY:~$ gcc ParentChild.c -o ParentChild
saicharan_reddi@SAICHARANREDDY:~$ ./ParentChild
Parent process started
Parent PID   : 735
Parent PPID  : 387
State       : Running

Parent process
Parent PID   : 735
Child PID    : 736
State       : Waiting
Child process created
Child PID    : 736
Child PPID   : 735
State       : Running
Child PID 736
State       : Terminated
Child process completed
Parent State: Running
Parent State: Terminated
saicharan_reddi@SAICHARANREDDY:~$ |

```

REPORT: The program successfully creates a child process using fork(). The PID and PPID of the parent and child processes are displayed, and the process states are observed during execution.

Task 4: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations

CODE:

PS -EF:

```
saicharan_reddi@SAICHARANREDDY:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1        0  0  10:30 ?        00:00:00 /sbin/init
root           2        1  0  10:30 hvc0     00:00:00 /init
root           6        2  0  10:30 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          46        1  0  10:30 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+      78        1  0  10:30 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root          80        1  0  10:30 ?        00:00:00 /usr/lib/systemd/systemd-udevd
root         161        1  0  10:30 ?        00:00:01 snapfuse /var/lib/snapd/snaps/snapd_27591.snap /snap/snapd/27591 -o ro,nodev,allow_other,suid
root         162        1  0  10:30 ?        00:00:00 snapfuse /var/lib/snapd/snaps/core22_2437.snap /snap/core22/2437 -o ro,nodev,allow_other,suid
root         163        1  0  10:30 ?        00:00:00 snapfuse /var/lib/snapd/snaps/core22_2411.snap /snap/core22/2411 -o ro,nodev,allow_other,suid
root         173        1  0  10:30 ?        00:00:00 snapfuse /var/lib/snapd/snaps/tree_54.snap /snap/tree/54 -o ro,nodev,allow_other,suid
root         197        1  0  10:30 ?        00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         198        1  0  10:30 ?        00:00:00 /usr/sbin/cron -f -P
message+     199        1  0  10:30 ?        00:00:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root         203        1  0  10:30 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         206        1  0  10:30 ?        00:00:00 /snap/snapd/current/usr/lib/snapd/snapd
root         207        1  0  10:30 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         209        1  0  10:30 ?        00:00:00 /usr/libexec/wsl-pro-service
syslog       228        1  0  10:30 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         230        1  0  10:30 tty1     00:00:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d
_chrony      252       197  0  10:30 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
root         256        1  0  10:30 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
_chrony      273       252  0  10:30 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
root         380        2  0  10:30 ?        00:00:00 /init
root         381       380  0  10:30 ?        00:00:00 /init
saichart+    387       381  0  10:30 pts/0    00:00:00 -bash
root         388        2  0  10:30 ?        00:00:00 login -- saicharan_reddi
saichart+   442        1  0  10:30 ?        00:00:00 /usr/lib/systemd/systemd --user
saichart+   444       442  0  10:30 ?        00:00:00 (sd-pam)
saichart+   472       388  0  10:30 pts/1    00:00:00 -bash
root         763        80  0  10:34 ?        00:00:00 (udev-worker)
root         764        80  0  10:34 ?        00:00:00 (udev-worker)
saichart+   770       387  0  10:34 pts/0    00:00:00 ps -ef
saicharan_reddi@SAICHARANREDDY:~$
```

PS:

```
saicharan_reddi@SAICHARANREDDY:~$ ps -o pid,ppid,state,cmd
PID      PPID S  CMD
387       381 S  -bash
788       387 R  ps -o pid,ppid,state,cmd
saicharan_reddi@SAICHARANREDDY:~$
```

TOP:

```
saicharan_reddi@SAICHARAN ~$ top
top - 10:36:33 up 6 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 32 total, 1 running, 31 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7826.6 total, 6045.4 free, 574.2 used, 1360.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7252.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    80 root        20   0   35672   12608   9228 S   0.3   0.2   0:00.53 systemd-udevd
     1 root        20   0   24132   15484  11528 S   0.0   0.2   0:00.52 systemd
     2 root        20   0    3180    2212   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
     6 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
    46 root        19  -1   50404   16744  15448 S   0.0   0.2   0:00.14 systemd-journal
    78 systemd+    20   0    22416   14480  11980 S   0.0   0.2   0:00.06 systemd-resolve
   161 root        20   0   380412  11920   1516 S   0.0   0.1   0:01.45 snapfuse
   162 root        20   0   159348   1880   1556 S   0.0   0.0   0:00.00 snapfuse
   163 root        20   0   85484    1484   1296 S   0.0   0.0   0:00.00 snapfuse
   173 root        20   0   159348   1844   1516 S   0.0   0.0   0:00.00 snapfuse
   197 root        20   0    2888    1948   1820 S   0.0   0.0   0:00.02 chronyd-starter
   198 root        20   0    4472    3144   2900 S   0.0   0.0   0:00.00 cron
   199 message+    20   0    8816    5308   4628 S   0.0   0.1   0:00.03 dbus-daemon
   203 root        20   0    44096   29804  14660 S   0.0   0.4   0:00.17 networkd-dispat
   206 root        20   0  2295248   38072  24268 S   0.0   0.5   0:00.49 snapd
   207 root        20   0   18440    9324   8080 S   0.0   0.1   0:00.05 systemd-logind
   209 root        20   0  1857908   15292  12616 S   0.0   0.2   0:00.07 wsl-pro-service
   228 syslog      20   0   220548    5916   4604 S   0.0   0.1   0:00.03 rsyslogd
   230 root        20   0    5492    2760   2580 S   0.0   0.0   0:00.00 agetty
   252 _chrony      20   0    20424    6332   5560 S   0.0   0.1   0:00.03 chronyd
   256 root        20   0   123460  32508  17900 S   0.0   0.4   0:00.15 unattended-upgr
   273 _chrony      20   0    12096    2440   1800 S   0.0   0.0   0:00.00 chronyd
   380 root        20   0    3184    1116    980 S   0.0   0.0   0:00.00 SessionLeader
   381 root        20   0    3200    1252   1108 S   0.0   0.0   0:00.08 Relay(387)
   387 saichar+    20   0    6268    5520   3848 S   0.0   0.1   0:00.04 bash
   388 root        20   0    8648    5592   4748 S   0.0   0.1   0:00.01 login
   442 saichar+    20   0   22356   13600  10964 S   0.0   0.2   0:00.10 systemd
   444 saichar+    20   0   22920    4124   2120 S   0.0   0.1   0:00.00 (sd-pam)
   472 saichar+    20   0    6136    5476   3928 S   0.0   0.1   0:00.03 bash
   815 saichar+    20   0    8172    6012   3888 R   0.0   0.1   0:00.14 top
   869 root        20   0   35676    6752   3336 S   0.0   0.1   0:00.00 (udev-worker)
   870 root        20   0   35676    6752   3336 S   0.0   0.1   0:00.00 (udev-worker)
```

REPORT: This is understanding process creation and process state transitions in Linux. The `fork()` system call was used to create a child process, while `ps`, `top`, and `/proc` were used to monitor process information and states. The PID and PPID helped identify the relationship between the parent and child processes.